Tactix: recovered SAT coverage and proofless result parity

Codex for Andrew

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Outcome. The repaired Tactix CLI can solve without producing a proof: call `tactix <cnf>` with no second path argument. Its full-width pinned Rank 1–50 diagnostic, SLURM array `1729452`, completed all 50 tasks with exit zero. All 50 verdicts match the historically checked AM337 proof-producing Rank reference: 25 SAT and 25 UNSAT, with no timeout. All 25 SAT assignments passed the pinned checker against the original CNF. The array used the same repaired Release binary SHA-256 `c57482f1551e48b6f460398790b605b75cb9149e96d38e8406e2b1f2b5315631`; its task results were checked against the immutable input and matrix hashes. The summed solver time was 7,207.560 seconds, with a 665.748-second maximum (Rank 34). This confirms result parity, not identical assignments, search trajectories, or an isolated performance comparison. Proofless UNSAT verdicts are not independently certified by that run.

Five lost SAT cases. The old-source automatic-proof Rank campaign finished 45/50 because Ranks 7, 8, 10, 13, and 14 each exceeded 1,200 seconds, despite an earlier checked 50/50 baseline. A 12,000-sample Rank 7 profile assigned 99.24% of native DSR user cycles to an internal-map scan. The CLI was missing the reverse external-variable-map binding already used by the API. Commit `bf2fbdcd` adds that binding without changing the public rule surface or proof-checker contract. On the repaired pin, the five-case automatic-proof job `1729413` and paired proofless job `1729420` both returned SAT on all five inputs, with every original-CNF model checked. Automatic-proof solve times were 88.232, 94.131, 162.512, 104.544, and 124.769 seconds; proofless times were 69.729, 89.729, 106.501, 110.030, and 91.433 seconds. These diagnostic jobs overlapped other cluster work, so timing parity with the historical baseline remains unproved.

Certificate and performance limits. A repaired-source targeted automatic-proof Rank 12 run produced a complete DRAT-to-CakeLPR certificate independently verified against the original CNF; its paired proofless run returned the same UNSAT verdict. Rank 5 SAT models were checked in both modes. Full repaired-source automatic-proof Rank and B28+C5 campaigns are prepared with the plan's seven-checker content route and original resource/time gates, but have not been submitted: latest-proof-source Opus fuzz AM473 has already found further native DSR rejections in the checker-retained-root-reason family. A general proof-state repair is required before rerunning the combined N2 and Phase 1 gates. The original public X7 actions and no-option native DSR-to-LSR-to-Trestle contract remain intact. The old-source Rank campaign's 45/50 and 19,148.160 PAR2 do not meet its 8,115.735-second gate; its B28+C5 arm checked 33/33 in 920.163 seconds. The full Debug job `1729059` was stopped after an outdated exact-byte binary-deletion fixture failed; the fixture now reflects the general split-unit proof transaction and passes focused Release and sanitizer Debug tests. Full current-source Debug remains due.

Next actions. Collect AM473's bounded failure and implement a general native DSR proof-state transaction without per-instance tuning or checker-option changes; then repeat the current-source fuzz, full Debug, independent certificate/model, Rank, B28+C5, WV100, and WV2-89 gates. Finish the producer-by-producer winner proof-rule audit and N1–N4 acceptance before the final online-learning readiness decision. Only then submit the 1,166-instance Tier D run at 5,000 seconds, 4 CPUs, and 32 GiB on `cpu-dense-hwulab-q`. It has not been submitted.

Evidence. `benchmarks/reports/2026-09-26-am475-five-sat-recovery.md`; `benchmarks/reports/2026-09-26-am474-rank7-dsr-map-bound.md`; `benchmarks/runs/20260926-am475-rank1-50-proofless-diagnostic/`; `benchmarks/runs/20260926-am473-batch-root-claude-fuzz/OWNER_STATUS.md`; `benchmarks/reports/2026-09-26-am471-automatic-proof-gate-contract.md`.